

Simplifying Surds

AQA · KS4 Year 10–11

Name: _____ Class: _____ Date: _____

Simplify surds by finding the largest square factor. Express surds in their simplest form $a\sqrt{b}$. Understand why surds are 'exact' values and when to leave answers in surd form.

VISUAL EXAMPLE

$$\begin{aligned}\sqrt{72} &= \sqrt{(36 \times 2)} \\ &= \sqrt{36} \times \sqrt{2} \\ &= 6\sqrt{2}\end{aligned}$$

Simplifying a surd: split off the largest square factor.

COMMON MISTAKES

- ✗ Not finding the LARGEST square factor — $\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$, not $\sqrt{(4 \times 18)} = 2\sqrt{18}$ (not fully simplified).
- ✗ Thinking $\sqrt{(a + b)} = \sqrt{a} + \sqrt{b}$ — this is WRONG. $\sqrt{(9 + 16)} \neq \sqrt{9} + \sqrt{16}$.
- ✗ Writing $\sqrt{4} \times \sqrt{3} = \sqrt{12}$ is correct, but the point of simplifying is to go the OTHER way: $\sqrt{12} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$.
- ✗ Forgetting that $\sqrt{a} \times \sqrt{a} = a$ (not $\sqrt{a^2}$).

METHOD

- 1 Find the largest perfect square that is a factor of the number under the $\sqrt{}$.
- 2 Split: $\sqrt{n} = \sqrt{(\text{square factor} \times \text{remainder})}$.
- 3 Use $\sqrt{(a \times b)} = \sqrt{a} \times \sqrt{b}$.
- 4 Take the square root of the perfect square factor out.
- 5 Write answer as $k\sqrt{m}$ where m has no square factors.

WORKED EXAMPLE

Simplify $\sqrt{72}$.

Step 1 — Largest square factor of 72: 36 (since $36 \times 2 = 72$).

Step 2 — $\sqrt{72} = \sqrt{(36 \times 2)}$

Step 3 — $= \sqrt{36} \times \sqrt{2}$

Step 4 — $= 6\sqrt{2}$

Check: $6^2 \times 2 = 36 \times 2 = 72$ ✓

1 Simplify: (2 marks)

a) $\sqrt{12}$ b) $\sqrt{18}$ c) $\sqrt{20}$ d) $\sqrt{50}$

HOW TO START

- Find the largest square number that divides into the number under the root (e.g. 4, 9, 16, 25, 36...).
- Split it: $\sqrt{12} = \sqrt{(4 \times 3)} = \sqrt{4} \times \sqrt{3}$.
- Take the square root of the square factor: $\sqrt{4} = 2$, so $\sqrt{12} = 2\sqrt{3}$.

ANSWERS

(a) = _____
(b) = _____
(c) = _____
(d) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{(a \times b)} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

2 Simplify:

(2 marks)

a) $\sqrt{8}$ b) $\sqrt{32}$ c) $\sqrt{45}$ d) $\sqrt{75}$

HOW TO START

→ Same method: largest square factor first.

→ For $\sqrt{8}$, the largest square factor is 4: $\sqrt{8} = \sqrt{4 \times 2} = 2\sqrt{2}$.

ANSWERS

(a) = _____

(b) = _____

(c) = _____

(d) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

3 Simplify: (3 marks)

- a) $\sqrt{98}$ b) $\sqrt{200}$ c) $\sqrt{128}$ d) $\sqrt{300}$

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____
- (d) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

4 Simplify: (3 marks)

- a) $\sqrt{4 \times 7}$ b) $3\sqrt{12}$ c) $2\sqrt{50}$ d) $5\sqrt{8}$

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____
- (d) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

5 Simplify fully:

(3 marks)

a) $\sqrt{48} + \sqrt{12}$ b) $\sqrt{75} - \sqrt{27}$ c) $2\sqrt{18} + 3\sqrt{2}$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

6 Show that:

(4 marks)

a) $\sqrt{80} = 4\sqrt{5}$ b) $3\sqrt{28} = 6\sqrt{7}$ c) $\sqrt{180} = 6\sqrt{5}$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

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(4 marks)

- a) Find the exact side length. (1 mark)

b) Find the exact perimeter. (1 mark)

c) A second square has side length $3\sqrt{2}$ cm. Which square is larger?

Show working. (2 marks)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

8 a) Show that $\sqrt{a^2b} = a\sqrt{b}$ for any positive a and b . (1 mark)

(5 marks)

b) Simplify $\sqrt{50x^4y^2}$ where $x, y > 0$. (2 marks)

c) A right-angled triangle has shorter sides $\sqrt{18}$ cm and $\sqrt{32}$ cm. Find the exact hypotenuse in simplest surd form. (2 marks)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find largest square factor | 2 Split under the root: $\sqrt{(\text{square} \times \text{rest})}$ | 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k\sqrt{m}$ (m has no square factors)

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet

☐ Getting there

☐ Confident

☐ Really confident

Which question did you find the hardest, and why?

Do I always find the LARGEST square factor first time?

One thing I will practise before the next lesson:

Teacher key

Q1: a) $2\sqrt{3}$ b) $3\sqrt{2}$ c) $2\sqrt{5}$ d) $5\sqrt{2}$

Q2: a) $2\sqrt{2}$ b) $4\sqrt{2}$ c) $3\sqrt{5}$ d) $5\sqrt{3}$

Q3: a) $7\sqrt{2}$ b) $10\sqrt{2}$ c) $8\sqrt{2}$ d) $10\sqrt{3}$

Q4: a) $2\sqrt{7}$ b) $3 \times 2\sqrt{3} = 6\sqrt{3}$ c) $2 \times 5\sqrt{2} = 10\sqrt{2}$ d) $5 \times 2\sqrt{2} = 10\sqrt{2}$

Q5: a) $4\sqrt{3} + 2\sqrt{3} = 6\sqrt{3}$ b) $5\sqrt{3} - 3\sqrt{3} = 2\sqrt{3}$ c) $6\sqrt{2} + 3\sqrt{2} = 9\sqrt{2}$

Q6: a) $\sqrt{80} = \sqrt{(16 \times 5)} = 4\sqrt{5}$ ✓ b) $3\sqrt{28} = 3\sqrt{(4 \times 7)} = 3 \times 2\sqrt{7} = 6\sqrt{7}$ ✓ c) $\sqrt{180} = \sqrt{(36 \times 5)} = 6\sqrt{5}$ ✓

Q7: a) $\sqrt{72} = 6\sqrt{2}$ cm b) $4 \times 6\sqrt{2} = 24\sqrt{2}$ cm c) Second square side = $3\sqrt{2}$, first = $6\sqrt{2}$. $6\sqrt{2} > 3\sqrt{2}$, so first is larger.

Q8: a) $\sqrt{(a^2b)} = \sqrt{(a^2)} \times \sqrt{b} = a\sqrt{b}$ ✓ b) $\sqrt{(50x^4y^2)} = \sqrt{(25 \times 2 \times x^4 \times y^2)} = 5x^2y\sqrt{2}$ c) $\text{Hyp}^2 = 18 + 32 = 50$; $\text{hyp} = \sqrt{50} = 5\sqrt{2}$ cm
